

EXPLORATORY COMPUTATIONAL PROJECT

Posterior Geometry in Variational Inference and Langevin Sampling

Structural Approximation and
Geometry-Dependent Mixing

Author

Katia Russo

Supervisor

Prof. Giacomo Zanella

Bocconi University

July 2026

Contents

Introduction	3
1 Mean-field variational inference as structural approximation	7
1.1 Variational inference as optimization over a tractable family	7
1.2 The mean-field restriction	8
1.3 Correlated Gaussian targets	10
1.4 Optimal mean-field Gaussian approximation	11
2 Langevin dynamics and geometry-dependent mixing	17
2.1 From posterior computation to Langevin dynamics	17
2.2 Continuous-time Langevin dynamics	18
2.3 ULA as discretization and its bias	21
2.4 Geometry-dependent complexity under log-concavity	22
3 Controlled experiments on posterior geometry	29
3.1 Purpose and methodology	29
3.2 Experiment 1: Bivariate Gaussian target	30
3.3 Experiment 2: Bayesian linear regression with collinear design	35

4	Discussion	47
4.1	The structural-dynamical asymmetry	47
4.2	Beyond the two prototypes	49
	Conclusion	53
A	Computational details	55

Introduction

Statistical modelling often starts from an incomplete view of a system. Data are observed, but the quantities that generated them are not: parameters, latent variables, or hidden states must be inferred from indirect evidence. A point estimate can summarize one plausible value, but it does not describe how uncertain that value is, how different unknown quantities interact, or how strongly the data identify them. In this sense, the inferential object of interest is not only a single parameter value, but a distribution over the unknowns after the data have been observed.

Bayesian inference provides a natural formalism for this task. Let x denote the unknown parameters or latent variables and let y denote the observed data. Starting from a joint model $p(x, y)$, the information contained in the observations is incorporated through the posterior distribution

$$p(x \mid y) = \frac{p(x, y)}{p(y)}, \quad p(y) = \int p(x, y) dx.$$

The posterior is therefore the central object of Bayesian inference: it combines the model assumptions and the observed data into a probability distribution over the unknown quantities.

In simple conjugate models this distribution is available analytically. In more realistic models, however, the normalizing constant $p(y)$ and the posterior expectations of interest are usually not available in closed form. The conceptual answer provided by Bayes' rule

then becomes a computational problem: the posterior is well defined, but it cannot be manipulated exactly and must be approximated.

Once exact posterior computation is no longer available, one has to decide what kind of approximation is acceptable. One possibility is to approximate the posterior indirectly, by generating samples from a Markov chain whose stationary distribution is the target posterior and then estimating expectations by ergodic averages. This is the sampling perspective, represented by Markov chain Monte Carlo methods. A different possibility is to replace the posterior by a tractable distribution chosen through optimization. Variational inference follows this second route: it turns posterior approximation into the problem of finding, within a prescribed family of distributions, the element closest to the posterior in Kullback-Leibler divergence (Bishop, 2006; Blei et al., 2017).

This work studies the contrast between these two approximation philosophies through the lens of posterior geometry. Here, posterior geometry means the structural features of the target distribution that help determine how difficult it is to approximate or explore. These features are not perceived in the same way by all methods. For a variational approximation, geometry matters primarily through representability: the chosen family may or may not be able to reproduce the dependence structure of the posterior. For a sampling algorithm, geometry matters through dynamics: the shape of the target determines how efficiently the algorithm can move through the state space.

This distinction becomes especially transparent in Gaussian settings. For a Gaussian posterior, the covariance matrix describes marginal uncertainty, dependence between coordinates, and the orientation of level sets, while the precision matrix describes the curvature of the negative log-density. Dependence, curvature, and conditioning are therefore linked through the same linear-algebraic structure. Gaussian targets are consequently a useful controlled setting: the same posterior geometry can be examined from the variational side, where dependence creates a representational obstruction, and from the sampling side, where curvature and conditioning affect the dynamics. Outside the Gaussian case, these notions need not coincide, so the two interpretations of geometry should be kept

conceptually distinct.

The thesis focuses on two simple prototypes. On the variational side, it studies fully factorized mean-field variational inference. Its limitation is structural: once independence is imposed by the approximating family, posterior dependence cannot be recovered merely by solving the optimization problem more accurately. On the sampling side, it studies the unadjusted Langevin algorithm. Here the limitation is of a different nature. The continuous Langevin diffusion has the posterior as invariant distribution under suitable assumptions, but the Euler discretization used in ULA introduces bias, while curvature anisotropy slows exploration through the condition number of the target (Roberts and Tweedie, 1996; Dalalyan, 2017b).

The following chapters develop this comparison in a controlled mathematical setting. Chapter 1 derives the optimal mean-field approximation to a correlated Gaussian target and shows that the approximation error is irreducible within the factorized family. Chapter 2 studies Langevin dynamics and ULA, separating the invariant law of the continuous diffusion from the bias and slowdown introduced by discretization. Chapter 3 uses bivariate Gaussian targets and Bayesian linear regression with collinear design to display both mechanisms in settings where the exact posterior is known. Chapter 4 discusses how the two failure modes should be interpreted and places the comparison in relation to richer variational families, improved sampling dynamics, and coordinate-wise methods. The conclusion summarizes the scope of the analysis and formulates the main methodological implication: posterior geometry is not a single scalar difficulty, but a method-dependent diagnostic.

Chapter 1

Mean-field variational inference as structural approximation

1.1 Variational inference as optimization over a tractable family

Variational inference addresses the intractability of the posterior by replacing exact posterior computation with an optimization problem over a restricted family of distributions. One posits a family $\mathcal{Q}$ of candidate approximations and seeks the element of that family that is closest to the target posterior in Kullback-Leibler divergence (Blei et al., 2017; Bishop, 2006).

The posterior distribution of interest is

$$p(x | y) = \frac{p(x, y)}{p(y)},$$

where $p(y) = \int p(x, y)dx$ is the marginal likelihood. In complex Bayesian models, this normalizing constant is typically intractable and direct optimization with respect to the posterior is inconvenient. Variational inference circumvents this difficulty by introducing a candidate distribution approximation $q(x) \in \mathcal{Q}$ and optimizing a surrogate objective. The key identity is:

$$\log p(y) = \mathcal{L}(q) + \text{KL}(q(x) \| p(x | y)),$$

where

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(x, y)] - \mathbb{E}_q[\log q(x)]$$

is the evidence lower bound (ELBO), and

$$\text{KL}(q(x) \| p(x | y)) = \mathbb{E}_q \left[\log \frac{q(x)}{p(x | y)} \right]$$

is the Kullback-Leibler divergence from q to the posterior. Since the KL divergence is nonnegative, the ELBO is indeed a lower bound on the log marginal likelihood.

Proposition 1. For any variational family $\mathcal{Q}$,

$$\arg \max_{q \in \mathcal{Q}} \mathcal{L}(q) = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(x) \| p(x | y)).$$

Proof. From the decomposition

$$\log p(y) = \mathcal{L}(q) + \text{KL}(q(x) \| p(x | y)),$$

and since $\log p(y)$ is constant with respect to q , maximizing $\mathcal{L}(q)$ is equivalent to minimizing $\text{KL}(q \| p(\cdot | y))$. □

Thus, variational inference can be viewed either as lower-bound maximization or as projection of the posterior onto a tractable family $\mathcal{Q}$.

The usefulness of this formulation depends entirely on the choice of $\mathcal{Q}$. A richer family may provide more accurate approximations but lead to a difficult optimization problem; a simpler family may yield a tractable objective at the price of structural approximation error.

1.2 The mean-field restriction

A classical way to obtain a tractable variational problem is to restrict the approximating family to distributions that factorize across coordinates or groups of variables. In its sim-

plest form, the mean-field approximation assumes that the latent variables are mutually independent under the variational distribution. If $x = (x_1, \dots, x_d)$, the fully factorized mean-field family is

$$\mathcal{Q}_{MF} = \left\{ q(x) : q(x) = \prod_{j=1}^d q_j(x_j) \right\}.$$

This is the standard mean-field family considered in variational inference (Blei et al., 2017; Bishop, 2006).

The appeal of this restriction is computational. By replacing a general joint distribution with a product form, the optimization is carried out over low-dimensional factors rather than over an unrestricted multivariate density. In many models, this factorization turns the ELBO into an objective that is analytically simpler and more amenable to iterative optimization. For this reason, mean-field variational inference is often the first approximation considered when exact posterior computation is infeasible.

This simplification, however, comes with a structural cost. The family $\mathcal{Q}_{MF}$ does not merely regularize or simplify the optimization problem; it rules out any posterior dependence between distinct coordinates. In particular, for every $q \in \mathcal{Q}_{MF}$, the joint law under q is determined entirely by its marginals, and all dependence is removed by construction. Consequently, if the true posterior $p(x \mid y)$ exhibits substantial dependence between coordinates, then this dependence structure lies outside the variational family itself. In such case, even the variational optimum within $\mathcal{Q}_{MF}$ may remain qualitatively different from the target posterior.

It is useful to emphasize the distinction between optimization error and approximation error. Optimization error arises when the ELBO is not solved accurately within the chosen family. Approximation error, by contrast, is present even when the ELBO is optimized exactly, provided that the posterior does not belong to the family over which one is optimizing. For the mean-field family, this second phenomenon is especially transparent: the restriction to independent factors can make the best in-family approximation fundamentally unable to represent the dependence structure of the true posterior.

The mean-field family is therefore not introduced as a generic algorithmic convenience, but as a mathematically explicit structural restriction.

1.3 Correlated Gaussian targets

To understand the consequences of the mean-field restriction, it is useful to work in a setting where the geometry of the target distribution is completely explicit. For this reason, we focus on correlated Gaussian posteriors. Let

$$p(x) = \mathcal{N}(\mu, \Sigma), \quad x \in \mathbb{R}^d,$$

where $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d \times d}$ is symmetric positive definite.

This choice is not meant to be a claim of generality. Rather, it is a deliberately controlled setting in which the relation between posterior geometry and variational approximation can be analysed exactly. In a Gaussian target, the mean vector μ determines location, while the covariance matrix Σ determines spread, orientation, and dependence structure. In particular, the off-diagonal entries of Σ encode correlation between coordinates, and the eigenstructure of Σ determines the principal axes and elongation of the level sets. Thus, in the Gaussian case, posterior geometry can be read directly from covariance geometry.

This makes correlated Gaussians the natural test case for the present thesis. If the variational family is fully factorized, then any approximation $q \in \mathcal{Q}_{MF}$ must suppress all posterior dependence by construction. When the target is Gaussian, this loss of dependence can be characterized analytically rather than only described qualitatively. The Gaussian setting therefore allows one to isolate structural approximation error from other sources of discrepancy.

A second advantage of this setting is methodological. Because both the target density and the variational family admit closed-form expressions, the Kullback-Leibler objective can be written explicitly and optimized exactly within the mean-field family. As a consequence,

the gap between the variational optimum and the true posterior cannot be attributed to poor numerical optimization. If a mismatch remains, it must come from the restriction imposed by the family itself.

For intuition, it is helpful to consider first the bivariate case,

$$p(x) = \mathcal{N} \left(\mu, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right), \quad |\rho| < 1.$$

When $\rho \neq 0$, the covariance matrix has nonzero off-diagonal entries, so the principal axes of the Gaussian level sets are generally not aligned with the coordinate axes. By contrast, a fully factorized Gaussian approximation can only produce axis-aligned level sets. Already in two dimensions, therefore, the geometric content of the mean-field restriction becomes visible: the family does not merely simplify the target, but excludes its correlation structure a priori.

1.4 Optimal mean-field Gaussian approximation

In this section we solve the mean-field variational problem explicitly for a Gaussian target. The purpose is to exhibit, in closed form, the structural mismatch induced by the factorization constraint. Let the target posterior be

$$p(x) = \mathcal{N}(\mu, \Sigma), \quad x \in \mathbb{R}^d,$$

where Σ is symmetric positive definite. In accordance with the mean-field restriction introduced above, we approximate p within the family of Gaussian densities with diagonal covariance,

$$q(x) = \mathcal{N}(m, D), \quad D = \text{diag}(d_1, \dots, d_d), \quad d_j > 0.$$

$$\mathcal{Q}_{MF}^G = \{ \mathcal{N}(m, D) : m \in \mathbb{R}^d, D = \text{diag}(d_1, \dots, d_d), d_j > 0 \}.$$

This is the natural Gaussian subfamily of the fully factorized mean-field class. The variational problem is therefore

$$q^* = \arg \min_{q \in \mathcal{Q}_{MF}^G} KL(q||p).$$

The direction of the divergence is the standard one used in mean-field variational inference. We first compute the objective explicitly.

Proposition 2. Let $q = \mathcal{N}(m, D)$, $p = \mathcal{N}(\mu, \Sigma)$ with D diagonal positive definite and Σ positive definite. Then

$$KL(q||p) = \frac{1}{2} \left[\text{tr}(\Sigma^{-1}D) + (m - \mu)^\top \Sigma^{-1}(m - \mu) - d + \log \frac{\det \Sigma}{\det D} \right].$$

Proof. By definition

$$KL(q||p) = \mathbb{E}_q \left[\log \frac{q(x)}{p(x)} \right] = \mathbb{E}_q[\log q(x)] - \mathbb{E}_q[\log p(x)].$$

Using the Gaussian log-density formulas, we have

$$\log q(x) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log \det D - \frac{1}{2} (x - m)^\top D^{-1} (x - m),$$

and

$$\log p(x) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma - \frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu).$$

Taking expectation under q , we obtain

$$\mathbb{E}_q[(x - m)^\top D^{-1} (x - m)] = \text{tr}(D^{-1} \mathbb{E}_q[(x - m)(x - m)^\top]) = \text{tr}(D^{-1}D) = d,$$

and

$$\mathbb{E}_q[(x - \mu)^\top \Sigma^{-1} (x - \mu)] = \text{tr}(\Sigma^{-1} \mathbb{E}_q[(x - \mu)(x - \mu)^\top]) = \text{tr}(\Sigma^{-1}D) + (m - \mu)^\top \Sigma^{-1}(m - \mu).$$

Substituting into the definition of the KL divergence yields

$$\text{KL}(q\|p) = \frac{1}{2} \left[\text{tr}(\Sigma^{-1}D) + (m - \mu)^\top \Sigma^{-1}(m - \mu) - d + \log \frac{\det \Sigma}{\det D} \right].$$

□

The variational optimum can now be characterized exactly.

Proposition 3. Let $\Lambda = \Sigma^{-1}$. The unique minimizer of $\text{KL}(q\|p)$ over Gaussian mean-field approximations $q = \mathcal{N}(m, D)$ with diagonal D is given by

$$m^* = \mu, \quad D^* = \text{diag} \left(\frac{1}{\Lambda_{11}}, \dots, \frac{1}{\Lambda_{dd}} \right).$$

Proof. Rewriting the expression derived above for KL in terms of Λ , one obtains:

$$\text{KL}(q\|p) = \frac{1}{2} \left[\text{tr}(\Lambda D) + (m - \mu)^\top \Lambda (m - \mu) - d + \log \frac{\det \Sigma}{\det D} \right].$$

The terms involving m and D separate. Since Λ is positive definite, the quadratic form $(m - \mu)^\top \Lambda (m - \mu)$ is uniquely minimized at $m^* = \mu$.

Now since $D = \text{diag}(d_1, \dots, d_d)$, then

$$\text{tr}(\Lambda D) = \sum_{j=1}^d \Lambda_{jj} d_j, \quad \log \det D = \sum_{j=1}^d \log d_j.$$

Hence, up to constants independent of D , minimizing the KL is equivalent to minimizing

$$\sum_{j=1}^d (\Lambda_{jj} d_j - \log d_j).$$

Each coordinate decouples. For fixed j , define

$$f_j(d_j) = \Lambda_{jj} d_j - \log d_j, \quad d_j > 0.$$

$$f'_j(d_j) = \Lambda_{jj} - \frac{1}{d_j}, \quad f''_j(d_j) = \frac{1}{d_j^2} > 0.$$

Thus, f_j is strictly convex and has a unique minimizer $d_j^* = \frac{1}{\Lambda_{jj}}$. Therefore,

$$D^* = \text{diag} \left(\frac{1}{\Lambda_{11}}, \dots, \frac{1}{\Lambda_{dd}} \right).$$

□

The mean-field optimum preserves the posterior mean exactly, but it destroys posterior dependence by construction. The optimal covariance is diagonal, and its entries are determined by the diagonal of the precision matrix rather than by the full covariance structure. Therefore, unless Σ is already diagonal, the exact posterior does not belong to the mean-field family, and the variational optimum differs from the target even when the optimization is solved exactly.

Remark 1 (Bivariate case). Specializing to $d = 2$ with

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{pmatrix}, \quad |\rho| < 1,$$

the precision matrix is

$$\Lambda = \Sigma^{-1} = \frac{1}{1 - \rho^2} \begin{pmatrix} \sigma_1^{-2} & -\rho (\sigma_1 \sigma_2)^{-1} \\ -\rho (\sigma_1 \sigma_2)^{-1} & \sigma_2^{-2} \end{pmatrix},$$

so that $\Lambda_{11} = \sigma_1^{-2}/(1 - \rho^2)$ and $\Lambda_{22} = \sigma_2^{-2}/(1 - \rho^2)$. Substituting into the previous proposition,

$$m^* = \mu, \quad D^* = \begin{pmatrix} \sigma_1^2(1 - \rho^2) & 0 \\ 0 & \sigma_2^2(1 - \rho^2) \end{pmatrix}.$$

The variational optimum has the same mean as the target, zero correlation, and marginal variances shrunk by the factor $1 - \rho^2$ relative to the marginals of p . As $|\rho|$ grows, the mismatch becomes stronger: in the presence of correlation, mean-field variational inference

systematically under-represents posterior variance.

A useful corollary is that the variational error is strictly positive whenever the target is genuinely correlated.

Corollary 4. If Σ is not diagonal, then the Gaussian mean-field optimum satisfies

$$\min_{q \in \mathcal{Q}_{MF}^G} KL(q \| p) > 0.$$

Proof. By Proposition 3, the unique minimizer over $\mathcal{Q}_{MF}^G$ is $q^* = \mathcal{N}(\mu, D^*)$, where D^* is diagonal. If $KL(q^* \| p) = 0$, then $q^* = p$ almost everywhere, which for Gaussian densities implies equality of their covariance matrices. This would require $D^* = \Sigma$. Since D^* is diagonal, this is impossible whenever Σ is not diagonal. Hence the minimum is strictly positive. $\square$

This makes the nature of the failure completely explicit: the error is not caused by poor optimization, but by the fact that the family excludes the dependence structure of the target.

Chapter 2

Langevin dynamics and geometry-dependent mixing

2.1 From posterior computation to Langevin dynamics

The sampling side starts from a different response to posterior intractability. Instead of replacing the posterior by a simpler distribution, one constructs a stochastic process whose long-run behaviour is governed by the target itself. It is then convenient to write the posterior density on $\mathbb{R}^d$ in the form

$$\pi(x) \propto e^{-U(x)},$$

where $U : \mathbb{R}^d \rightarrow \mathbb{R}$ is the potential, or negative log-density up to an additive constant. This representation makes the connection between posterior geometry and algorithmic complexity explicit: the gradient ∇U determines the drift of the dynamics, while the Hessian $\nabla^2 U$ describes local curvature and conditioning (Dalalyan, 2017b; Chewi, 2026).

The computational task is no longer to approximate π by a tractable family, as in mean-field variational inference, but to construct a stochastic process whose invariant distribution is π , under suitable assumptions. Among gradient-based sampling methods, the canonical continuous-time object is the overdamped Langevin diffusion. Its importance comes from the fact that it combines two effects: a deterministic drift term that moves

the process toward regions of lower potential, and a Brownian noise term that makes the dynamics non-degenerate and, together with the drift, leads to π as invariant law under suitable conditions. In this sense, Langevin dynamics provides the natural sampling analogue of gradient-based optimization (Chewi, 2026; Dalalyan, 2017a).

This viewpoint is particularly suitable for the present thesis. On the variational side, posterior geometry enters through representability: a factorized family cannot encode dependence. On the sampling side, by contrast, the target remains in principle accessible, but the geometry of U affects how efficiently the dynamics can explore the state space. In targets with curvature anisotropy, directions of high curvature and directions of low curvature coexist, and this mismatch is precisely what will later translate into geometry-dependent slowdown for Langevin-based methods.

The continuous-time diffusion isolates this geometric mechanism in its cleanest form: π appears directly as the invariant law of an SDE, and any further error arises only when one passes to a discrete-time approximation.

2.2 Continuous-time Langevin dynamics

The canonical continuous-time process associated with the target density

$$\pi(x) \propto e^{-U(x)}, \quad x \in \mathbb{R}^d,$$

is the overdamped Langevin diffusion

$$dX_t = -\nabla U(X_t) dt + \sqrt{2} dB_t,$$

where $(B_t)_{t \geq 0}$ is a standard d -dimensional Brownian motion. This diffusion is the natural continuous-time benchmark for gradient-based sampling, and it is the object from which the unadjusted Langevin algorithm will later be obtained by Euler discretization (Chewi, 2026; Dalalyan, 2017b).

The reason this process is fundamental is that its drift is directly determined by the geometry of the target. Regions where U is steep induce a stronger deterministic pull, while flatter regions produce weaker drift. The noise term prevents the dynamics from reducing to deterministic gradient flow and, under suitable assumptions, allows the process to admit the full target distribution as invariant law rather than collapsing to a minimizer. In this sense, Langevin dynamics is the sampling analogue of gradient flow, but with stochastic forcing added so as to preserve a non-degenerate invariant law (Chewi, 2026; Dalalyan, 2017a).

For the purposes of this thesis, the key structural fact is that π is an invariant distribution of the continuous-time diffusion. This is the point at which the sampling side fundamentally differs from mean-field variational inference: in continuous time, the target is not replaced by a restricted family, but appears directly as the equilibrium law of the dynamics.

To state this precisely, let

$$\pi(x) = Z^{-1}e^{-U(x)}, \quad Z = \int_{\mathbb{R}^d} e^{-U(x)} dx < \infty,$$

and assume that $U \in C^2(\mathbb{R}^d)$ and that π is a probability density. Under standard regularity and integrability conditions, the Langevin diffusion has infinitesimal generator

$$\mathcal{L}f(x) = -\nabla U(x) \cdot \nabla f(x) + \Delta f(x),$$

for smooth test functions f .

Proposition 5. Assume $U \in C^2(\mathbb{R}^d)$ and $Z = \int_{\mathbb{R}^d} e^{-U(x)} dx < \infty$ and the usual regularity assumptions ensuring that the overdamped Langevin diffusion is well posed. Then $\pi(x) = Z^{-1}e^{-U(x)}$ is a stationary density for the diffusion.

Proof. The Fokker-Planck equation associated with the diffusion is

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla U) + \Delta \rho_t,$$

where ρ_t denotes the density of X_t , provided it exists and is sufficiently regular.

To check invariance, it is enough to verify that $\rho = \pi$ is a stationary solution, that is,

$$\nabla \cdot (\pi \nabla U) + \Delta \pi = 0.$$

Now,

$$\pi(x) = Z^{-1} e^{-U(x)} \implies \nabla \pi = -\pi \nabla U.$$

Since $\nabla \pi = -\pi \nabla U$, we have

$$\Delta \pi = \nabla \cdot (\nabla \pi) = -\nabla \cdot (\pi \nabla U).$$

Substituting this identity into the stationary Fokker-Planck equation yields

$$\nabla \cdot (\pi \nabla U) + \Delta \pi = \nabla \cdot (\pi \nabla U) - \nabla \cdot (\pi \nabla U) = 0$$

Hence, π is a stationary density for the diffusion. □

This proposition establishes the fundamental property of the continuous-time process: the target distribution is not approximated by a restricted family, but appears directly as the invariant law of the dynamics. The geometric difficulty is therefore not one of representability, but one of exploration: the process is in principle targeting the correct distribution, yet the shape of U determines how efficiently the state space is traversed.

The proposition, however, concerns the continuous-time diffusion, not an implementable algorithm. In general, the SDE cannot be simulated exactly and is therefore replaced by a discrete-time scheme, whose invariant law is no longer guaranteed to coincide with π (Dalalyan, 2017b; Roberts and Tweedie, 1996).

2.3 ULA as discretization and its bias

Since the continuous-time Langevin diffusion is not directly implementable, one replaces the stochastic differential equation with a discrete-time recursion. The most basic choice is the explicit Euler–Maruyama scheme, which leads to the unadjusted Langevin algorithm (ULA):

$$X_{k+1} = X_k - \eta \nabla U(X_k) + \sqrt{2\eta} \xi_k, \quad \xi_k \sim \mathcal{N}(0, I_d),$$

where $\eta > 0$ is the step size and $(\xi_k)_{k \geq 0}$ are i.i.d. standard Gaussian vectors. Thus ULA is obtained by replacing the continuous-time increment of the diffusion with a first-order discretization (Dalalyan, 2017b; Chewi, 2026).

At first sight, the recursion above may look like a straightforward implementable counterpart of the continuous diffusion. However, the passage from continuous to discrete time is not innocuous. The invariant distribution of the diffusion is the target π , but the Markov chain generated by ULA does not, in general, preserve π exactly. Instead, for fixed step size η , the chain may converge to a different stationary law, when such a law exists. This discrepancy is the discretization bias of ULA (Roberts and Tweedie, 1996).

On the continuous side, the target enters as the exact invariant measure of the dynamics. On the discrete side, the algorithm is only an approximation of that ideal process. Therefore, when analysing ULA, one must conceptually separate two effects. The first is the contraction of the dynamics toward equilibrium. The second is the error introduced by the fact that the equilibrium of the discretized chain is not exactly π . In the smooth and strongly log-concave regime, these two effects can both be controlled quantitatively, but they should not be conflated.

It is useful to make the analogy with optimization explicit. If one removes the Gaussian noise term, the recursion becomes

$$x_{k+1} = x_k - \eta \nabla U(x_k),$$

which is exactly gradient descent on the potential U . ULA can therefore be interpreted as the stochastic counterpart of gradient descent: the drift term pushes the iterate downhill, while the Gaussian perturbation prevents collapse to a minimizer and instead spreads the trajectory around the target distribution. This analogy is one of the main reasons why Langevin-based methods are analytically tractable and why they fit naturally within a geometry-driven comparison with variational methods (Dalalyan, 2017a).

At the same time, the analogy must not be overstated. Gradient descent converges to a minimizer of U , whereas ULA is intended to approximate a full probability distribution. This is exactly where discretization matters: the long-run law of ULA is not, in general, the target π , but a biased approximation whose quality depends on the step size η and on the geometry of the potential.

Metropolis-corrected variants such as the Metropolis-adjusted Langevin algorithm (MALA) restore π -invariance by adding an acceptance–rejection correction to the Euler proposal (Roberts and Tweedie, 1996). They do so at the cost of a more complicated algorithmic structure and of an acceptance probability that itself depends on the geometry of the target. ULA remains the correct object for the present thesis because the geometric effect we want to isolate is most transparent in the unadjusted recursion: the target is approached through a gradient-driven dynamics, but the price of discretization remains visible rather than being corrected away.

2.4 Geometry-dependent complexity under log-concavity

To make the role of geometry quantitative, it is standard to work in the strongly log-concave regime. Assume that

$$\pi(x) \propto e^{-U(x)}, \quad x \in \mathbb{R}^d,$$

where $U \in C^2(\mathbb{R}^d)$ satisfies the uniform Hessian bounds

$$mI_d \preceq \nabla^2 U(x) \preceq LI_d \quad \forall x \in \mathbb{R}^d,$$

for some constants $0 < m \leq L < \infty$. Equivalently, U is m -strongly convex and has L -Lipschitz gradient. In this setting, the ratio

$$\kappa = \frac{L}{m}$$

is the condition number of the target. It measures the degree of curvature anisotropy of the potential: $\kappa \approx 1$ corresponds to nearly isotropic geometry, whereas large κ corresponds to highly elongated level sets. This is the canonical framework for the non-asymptotic analysis of gradient-based sampling methods (Dalalyan, 2017b; Chewi, 2026; Durmus and Moulines, 2016).

A key point in this regime is that non-asymptotic bounds for ULA separate two different effects: a transient contraction toward equilibrium and a discretization error due to the Euler scheme. A standard non-asymptotic Wasserstein bound, with notation adapted from Dalalyan's M to the present L , has the schematic form

$$W_2(\mu_k, \pi) \leq (1 - m\eta)^k W_2(\mu_0, \pi) + C \frac{L}{m} \sqrt{\eta d},$$

(Dalalyan, 2017a, Theorem 1(a))¹ under the step-size restrictions of the theorem, where $C > 0$ is a universal constant and μ_k denotes the law of the k -th ULA iterate

$$X_{k+1} = X_k - \eta \nabla U(X_k) + \sqrt{2\eta} \xi_k, \quad \xi_k \sim \mathcal{N}(0, I_d).$$

The first term captures the transient contraction of the chain, while the second term is the price of the Euler discretization. Thus, for fixed η , ULA is not an exact sampler for π :

¹The precise theorem includes additional constants and step-size restrictions; the displayed form is used here only to emphasize the separation between transient contraction and discretization error.

it can approach equilibrium, but the equilibrium it approaches is generally biased. The bound makes this separation explicit. ULA is contractive, but not exact.

This non-asymptotic bound already shows why conditioning matters. If one chooses the step size at the natural scale $\eta \asymp L^{-1}$, then the contractive factor becomes

$$1 - m\eta \asymp 1 - \frac{m}{L} = 1 - \frac{1}{\kappa}.$$

Thus, large condition number implies slow contraction. At the same time, in this bound, the discretization term is amplified by the factor $L/m = \kappa$. In other words, anisotropy affects both parts of the error decomposition: it slows down the transient approach to equilibrium and worsens the bias induced by discretization.

Ignoring the discretization term for the moment, the same contraction factor gives the basic iteration-complexity interpretation. If η is chosen at the natural scale L^{-1} , then reducing the contractive term below a tolerance ε requires a number of iterations of order

$$O\left(\kappa \log \frac{1}{\varepsilon}\right),$$

up to constants and initial-distance terms. Thus the condition number enters directly as an iteration-complexity parameter. The discretization term then imposes an additional constraint on the step size when one wants the invariant-measure bias to be small.

The Gaussian case makes this mechanism completely explicit. Let

$$\pi = \mathcal{N}(0, \Sigma), \quad \Sigma \succ 0.$$

Then

$$U(x) = \frac{1}{2}x^\top \Sigma^{-1}x, \quad \nabla U(x) = \Sigma^{-1}x.$$

For a Gaussian target, the covariance matrix Σ and the precision matrix Σ^{-1} carry the same eigendirections with reciprocal eigenvalues. Thus the dependence structure de-

scribed by covariance geometry and the curvature structure described by the Hessian are linked through the same linear-algebraic structure. This coincidence is what will allow Chapter 3 to exhibit both failure modes simultaneously on a single target. The ULA recursion therefore becomes $X_{k+1} = (I - \eta\Sigma^{-1})X_k + \sqrt{2\eta}\xi_k$. If we denote $A_\eta = I - \eta\Sigma^{-1}$, then the chain is linear Gaussian:

$$X_{k+1} = A_\eta X_k + \sqrt{2\eta}\xi_k.$$

This allows the dynamics to be analysed in closed form.

Proposition 6. Let $\pi = \mathcal{N}(0, \Sigma)$ with $\Sigma \succ 0$, and consider the ULA recursion

$$X_{k+1} = (I - \eta\Sigma^{-1})X_k + \sqrt{2\eta}\xi_k, \quad \xi_k \sim \mathcal{N}(0, I_d).$$

Then the following hold:

- i. If $0 < \eta < \frac{2}{\lambda_{\max}(\Sigma^{-1})}$, then the linear Gaussian chain is stable, in the sense that the spectral radius of $I - \eta\Sigma^{-1}$ is strictly smaller than one. In particular, it admits a unique invariant Gaussian distribution.
- ii. If $\Sigma = Q\text{diag}(\nu_1, \dots, \nu_d)Q^\top$ with $\nu_i > 0$, and $\lambda_i = \nu_i^{-1}$, then in the rotated coordinates $Y_k = Q^\top X_k$, each component evolves independently as

$$Y_{k+1}^{(i)} = (1 - \eta\lambda_i)Y_k^{(i)} + \sqrt{2\eta}\zeta_k^{(i)}, \quad \zeta_k^{(i)} \sim \mathcal{N}(0, 1).$$

- iii. The stationary variance along eigendirection i is $c_\infty^{(i)} = \frac{\nu_i}{1 - \eta/(2\nu_i)}$, and therefore differs from the target variance ν_i whenever $\eta > 0$.

In particular, the invariant distribution of ULA is not equal to π for any positive step size.

Proof. We begin with the spectral decomposition, from which all three claims follow. Let

$$\Sigma = Q \operatorname{diag}(\nu_1, \dots, \nu_d) Q^\top, \quad \Sigma^{-1} = Q \operatorname{diag}(\lambda_1, \dots, \lambda_d) Q^\top, \quad \lambda_i = \nu_i^{-1}.$$

Since $A_\eta := I - \eta \Sigma^{-1}$ is a polynomial in Σ^{-1} , it is diagonalized by the same orthogonal matrix Q . Writing

$$Y_k = Q^\top X_k,$$

the recursion becomes

$$Y_{k+1} = (I - \eta \operatorname{diag}(\lambda_1, \dots, \lambda_d)) Y_k + \sqrt{2\eta} \zeta_k, \quad \zeta_k \sim \mathcal{N}(0, I_d).$$

Hence each coordinate evolves independently as

$$Y_{k+1}^{(i)} = (1 - \eta \lambda_i) Y_k^{(i)} + \sqrt{2\eta} \zeta_k^{(i)}, \quad \zeta_k^{(i)} \sim \mathcal{N}(0, 1),$$

which proves claim (ii).

For claim (i), the chain is stable if and only if $|1 - \eta \lambda_i| < 1 \ \forall i$, which is equivalent to

$$0 < \eta < \frac{2}{\lambda_{\max}(\Sigma^{-1})}.$$

For claim (iii), let

$$c_k^{(i)} := \operatorname{Var}(Y_k^{(i)}).$$

Using the independence of $Y_k^{(i)}$ and $\zeta_k^{(i)}$, we obtain

$$c_{k+1}^{(i)} = (1 - \eta \lambda_i)^2 c_k^{(i)} + 2\eta.$$

At stationarity, $c_{k+1}^{(i)} = c_k^{(i)} = c_\infty^{(i)}$, so

$$c_\infty^{(i)} = (1 - \eta \lambda_i)^2 c_\infty^{(i)} + 2\eta.$$

Hence,

$$c_{\infty}^{(i)} = \frac{2\eta}{1 - (1 - \eta\lambda_i)^2} = \frac{1}{\lambda_i - \frac{\eta}{2}\lambda_i^2} = \frac{\nu_i}{1 - \eta/(2\nu_i)}.$$

Under the stability condition, one has $0 < \eta < 2\nu_i$, hence

$$\frac{\nu_i}{1 - \eta/(2\nu_i)} > \nu_i.$$

Therefore the stationary variance along each eigendirection differs from the corresponding target variance. Hence, in this Gaussian example, the invariant distribution of ULA is not equal to π for any fixed $\eta > 0$. $\square$

This Gaussian calculation makes the geometric mechanism explicit. The contraction factor in direction i is $|1 - \eta\lambda_i|$. The largest-curvature direction, corresponding in the Gaussian case to $\lambda_{\max}(\Sigma^{-1}) = L$, imposes the stability restriction

$$\eta < \frac{2}{L}.$$

Once the step size is chosen small enough to control that stiff direction, the smallest-curvature direction, corresponding in the Gaussian case to $\lambda_{\min}(\Sigma^{-1}) = m$, contracts only at rate

$$|1 - \eta m|.$$

If η is chosen on the natural stability scale $\eta \asymp L^{-1}$, this becomes approximately

$$1 - \frac{m}{L} = 1 - \frac{1}{\kappa}.$$

Thus, the dynamics is forced to move cautiously because of the steep directions, and this caution makes progress slow in the flat directions. This is the geometric content of conditioning in Langevin sampling: anisotropy appears as dynamical stiffness.

The strongly log-concave regime therefore isolates a clean mechanism. The continuous-time diffusion has the correct invariant law, while its Euler discretization retains the same

geometric drift structure but introduces bias. The transient slowdown is governed by the conditioning of the potential, and the discretization error interacts with the same curvature parameters m and L . In the Gaussian case, both effects can be read directly from the eigendirections of Σ .

Chapter 3

Controlled experiments on posterior geometry

3.1 Purpose and methodology

The purpose of the experiments in this chapter is not to compare mean-field variational inference and ULA as black-box algorithms, but to isolate the geometric mechanisms identified above in settings where the exact target distribution is known. This makes it possible to distinguish structural approximation error from sampling inefficiency and discretization effects.

The first experiment uses a bivariate Gaussian target, while the second embeds the same type of geometry in a conjugate Bayesian linear regression model. In this way, both the optimal Gaussian mean-field approximation and the ULA dynamics can be analysed explicitly. This allows the representational failure of mean-field variational inference and the geometry-dependent slowdown of ULA to be visualized in the same example.

All deterministic quantities reported in the tables are computed from the closed-form Gaussian formulas derived in the previous chapters. Python is used to generate the figures and to simulate ULA trajectories for the empirical autocorrelation and effective-sample-size diagnostics. The computational details are reported in Appendix A.

3.2 Experiment 1: Bivariate Gaussian target

Consider the family of bivariate Gaussian target distributions

$$\pi_\rho = \mathcal{N}(0, \Sigma_\rho), \quad \Sigma_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \rho \in (-1, 1).$$

In the experiment, the values

$$\rho \in \{0, 0.5, 0.9, 0.99\}$$

are used. The parameter ρ controls the dependence between the two coordinates. When $\rho = 0$, the target is already factorized. As ρ approaches one, the level sets of the target become increasingly elongated and the two coordinates become increasingly dependent.

This example is useful because, in the Gaussian case, covariance structure and curvature are directly linked. The negative log-density is

$$U_\rho(x) = \frac{1}{2}x^\top \Sigma_\rho^{-1}x + \text{constant},$$

so the Hessian of the potential is the precision matrix

$$\nabla^2 U_\rho(x) = \Sigma_\rho^{-1}.$$

The eigenvalues of Σ_ρ are $1 + \rho$ and $1 - \rho$. Hence the spectral condition number is

$$\kappa = \frac{1 + \rho}{1 - \rho}.$$

Thus, increasing ρ simultaneously increases the dependence structure that is problematic for mean-field variational inference and the curvature anisotropy that affects Langevin dynamics. The two notions should not be identified in general, but in this Gaussian setting they can be studied side by side.

3.2.1 Mean-field approximation

The optimal Gaussian mean-field approximation derived in Chapter 1 has the form

$$q_\rho^* = \mathcal{N}(0, D_\rho^*),$$

where

$$D_\rho^* = \text{diag} \left(\frac{1}{\Lambda_{11}}, \frac{1}{\Lambda_{22}} \right), \quad \Lambda = \Sigma_\rho^{-1}.$$

For the present target,

$$\Sigma_\rho^{-1} = \frac{1}{1 - \rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix},$$

and therefore

$$D_\rho^* = (1 - \rho^2)I_2.$$

The optimal mean-field approximation has zero correlation for all values of ρ . Moreover, its marginal variances are shrunk from 1 to $1 - \rho^2$. The optimal KL divergence is

$$\text{KL}(q_\rho^* \parallel \pi_\rho) = -\frac{1}{2} \log(1 - \rho^2).$$

This quantity is zero when $\rho = 0$, and increases as ρ approaches one.

Table 3.1 reports the theoretical quantities used in the experiment. The results show the expected pattern: as the correlation increases, the condition number increases, the mean-field variance shrinks, and the optimal KL divergence becomes larger.

ρ	κ	η	D_ρ^* marginal variance	$\text{KL}(q_\rho^* \parallel \pi_\rho)$
0.00	1.0	0.800	1.0000	0.0000
0.50	3.0	0.400	0.7500	0.1438
0.90	19.0	0.080	0.1900	0.8304
0.99	199.0	0.008	0.0199	1.9585

Table 3.1: Theoretical quantities for the bivariate Gaussian experiment.

Figure 3.1 compares the contours of the target distribution and of the optimal Gaussian mean-field approximation. The solid contours represent π_ρ , while the dashed contours represent q_ρ^* . When $\rho = 0$, the target is already factorized and the mean-field approximation coincides with it. As ρ increases, the target becomes increasingly elongated and rotated, whereas the mean-field approximation remains axis-aligned and loses all dependence. This visualizes the structural nature of the approximation error: the mismatch remains even though the variational optimum is known exactly.

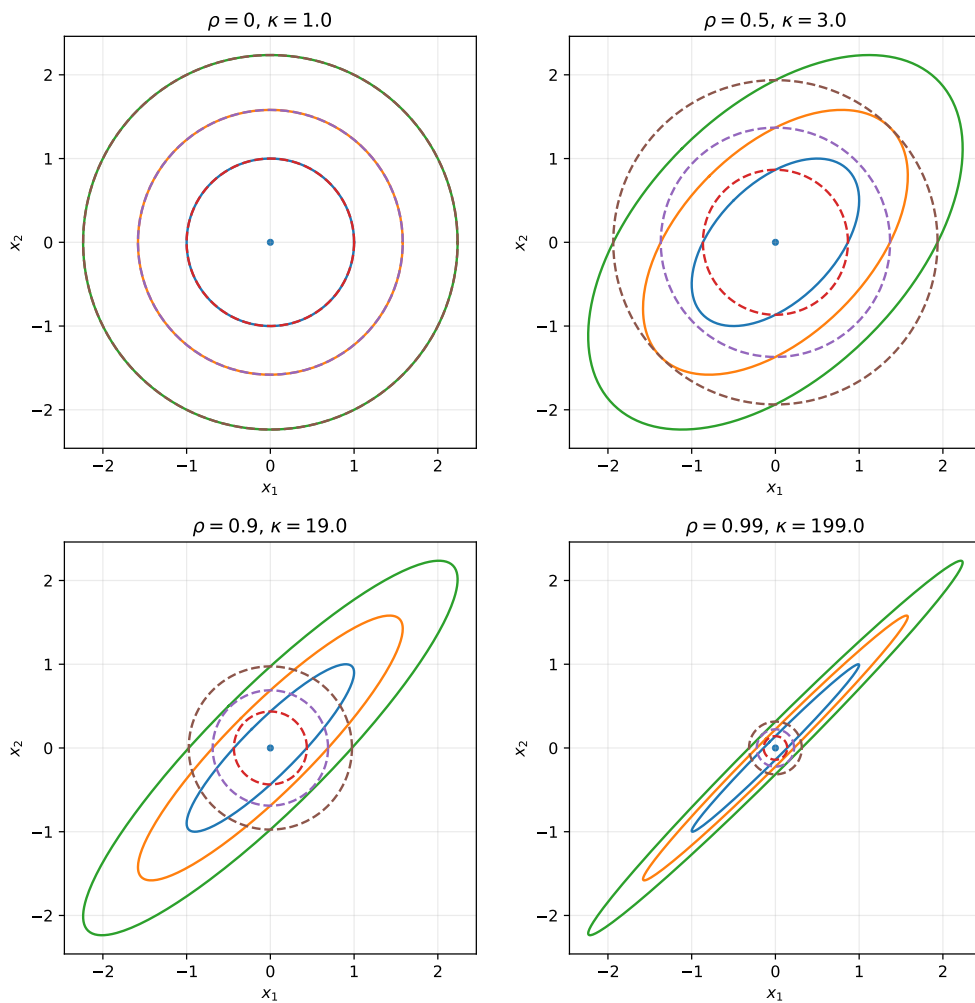


Figure 3.1: Target contours (solid) and optimal Gaussian mean-field approximations (dashed) for $\rho \in \{0, 0.5, 0.9, 0.99\}$.

3.2.2 ULA dynamics

The same target is then used to study the behaviour of ULA. Since

$$\nabla U_\rho(x) = \Sigma_\rho^{-1}x,$$

the ULA recursion is

$$X_{k+1} = X_k - \eta \Sigma_\rho^{-1} X_k + \sqrt{2\eta} \xi_k, \quad \xi_k \sim \mathcal{N}(0, I_2).$$

The step size is chosen as

$$\eta = \frac{0.8}{L}, \quad L = \lambda_{\max}(\Sigma_\rho^{-1}).$$

This choice keeps the chain inside the stability region for every value of ρ , while preserving the dependence of the slow direction on the condition number.

To analyse mixing, the samples are rotated into the eigenbasis of Σ_ρ . In this basis, the ULA chain decomposes into independent autoregressive processes. If λ_i is an eigenvalue of Σ_ρ^{-1} , then along the corresponding eigendirection

$$Y_{k+1}^{(i)} = (1 - \eta \lambda_i) Y_k^{(i)} + \sqrt{2\eta} \zeta_k^{(i)}.$$

The slow direction corresponds to the smallest curvature

$$m = \lambda_{\min}(\Sigma_\rho^{-1}).$$

With $\eta = 0.8/L$, the autoregressive coefficient in the slow direction is

$$1 - \eta m = 1 - \frac{0.8}{\kappa}.$$

Therefore, as κ increases, this coefficient approaches one and the autocorrelation decays

more slowly.

This behaviour is visible in Figure 3.2. For small ρ , the autocorrelation along the slow direction decays rapidly. For $\rho = 0.99$, where $\kappa = 199$, the autocorrelation remains high even at large lags. Anisotropy does not exclude the target from the sampling procedure; it makes exploration dynamically slow.

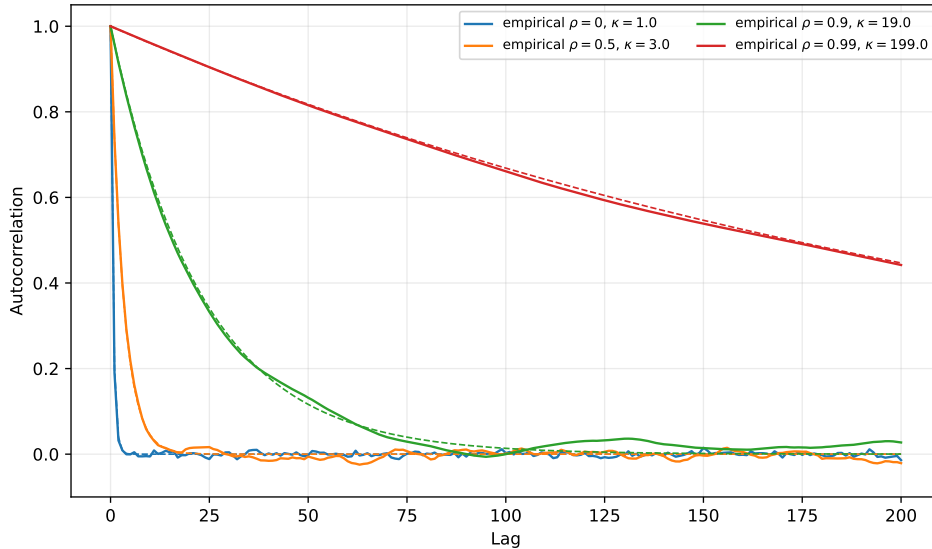


Figure 3.2: Empirical (solid) and theoretical AR(1) (dashed) autocorrelation of ULA along the slow eigendirection of Σ_ρ^{-1} .

The same conclusion is reflected in the effective sample size. Table 3.2 reports diagnostics computed from the simulated chains after burn-in. The effective sample size in the slow direction decreases sharply as ρ increases, while the fast direction remains comparatively well mixed. This directional difference is exactly what one expects from the spectral decomposition of the Gaussian target.

ρ	κ	η	ESS slow	ESS fast	ACF slow lag 100
0.00	1.0	0.800	27009.5	27366.2	0.0048
0.50	3.0	0.400	6182.9	26632.0	-0.0039
0.90	19.0	0.080	737.0	26417.5	0.0186
0.99	199.0	0.008	70.7	26444.1	0.6651

Table 3.2: ULA diagnostics for the bivariate Gaussian target. The effective sample size in the slow direction decreases as the condition number increases.

Finally, the Gaussian setting also makes the discretization bias of ULA explicit. For a fixed positive step size, the invariant distribution of ULA is not exactly the target. Along eigendirection i , if ν_i is the corresponding eigenvalue of Σ_ρ , the stationary variance of ULA is

$$c_\infty^{(i)} = \frac{\nu_i}{1 - \eta/(2\nu_i)}.$$

This differs from ν_i whenever $\eta > 0$. The purpose of this calculation is not to claim that the bias increases monotonically with ρ under the present step-size rule, but to show that discretization introduces an additional source of error absent from the continuous-time diffusion.

Thus, the same Gaussian target exhibits two distinct effects: a structural mismatch for the mean-field approximation and slow, biased dynamics for ULA.

3.3 Experiment 2: Bayesian linear regression with collinear design

The previous experiment imposed posterior geometry directly by choosing a correlated Gaussian target. The present experiment shows how the same geometry can arise from a standard Bayesian model. We consider Gaussian linear regression with a collinear design, where posterior dependence and anisotropy are induced by the likelihood rather than specified directly through a covariance matrix.

The experiment remains deliberately controlled. Since the model is conjugate, the posterior is available in closed form. This means that the quantities reported below are not estimates of an unknown ground truth: the posterior covariance, the mean-field optimum, the KL divergence, and the ULA eigenspace dynamics can all be computed from explicit formulas.

3.3.1 Model and posterior geometry

Consider the Gaussian linear regression model

$$y \mid \beta \sim \mathcal{N}(X\beta, \sigma^2 I_n), \quad \beta \sim \mathcal{N}(0, \tau^2 I_2),$$

where $X \in \mathbb{R}^{n \times 2}$, $\beta = (\beta_1, \beta_2)^\top$, and $\sigma^2, \tau^2 > 0$. Since both likelihood and prior are Gaussian, the posterior is also Gaussian:

$$\beta \mid y \sim \mathcal{N}(\mu_{\text{post}}, \Sigma_{\text{post}}),$$

with

$$\Sigma_{\text{post}} = \left(\frac{1}{\sigma^2} X^\top X + \frac{1}{\tau^2} I_2 \right)^{-1}, \quad \mu_{\text{post}} = \Sigma_{\text{post}} \frac{1}{\sigma^2} X^\top y.$$

Equivalently, the posterior precision matrix is

$$\Lambda_{\text{post}} = \Sigma_{\text{post}}^{-1} = \frac{1}{\sigma^2} X^\top X + \frac{1}{\tau^2} I_2.$$

This formula is the key object in the experiment. It shows that posterior geometry is determined by two components: the likelihood geometry, encoded by $X^\top X$, and the prior regularization, encoded by $\tau^{-2} I_2$.

The mean-field optimum derived in Chapter 1 is valid in arbitrary dimension. We set $p = 2$ not because the phenomenon is intrinsically bivariate, but because this is the only case in which the exact posterior, the mean-field approximation, and the geometry of their contours can be displayed simultaneously. To isolate the effect of design collinearity, the

design matrix is constructed so that

$$X^\top X = n \begin{pmatrix} 1 & \rho_X \\ \rho_X & 1 \end{pmatrix}, \quad \rho_X \in \{0, 0.5, 0.9, 0.99\}.$$

Thus ρ_X controls the empirical collinearity of the two regressors. This controlled construction is not meant to mimic a random design realization; rather, it removes accidental variability in $X^\top X$ and makes the dependence of posterior geometry on collinearity analytically explicit.

In the experiment we set

$$n = 100, \quad \sigma = 1, \quad \tau \in \{10, 1, 0.3\}.$$

The three values of τ represent a vague prior, a moderate prior, and a more informative prior. For the numerical visualizations involving the posterior mean, the data are generated as

$$y = X\beta_{\text{true}} + \sigma\varepsilon, \quad \beta_{\text{true}} = (1, -1)^\top, \quad \varepsilon \sim \mathcal{N}(0, I_n),$$

using a fixed random seed. The same noise vector is used across the different values of ρ_X and τ . This choice affects the posterior mean and therefore the location of the contour plots, but it does not affect the posterior covariance, the posterior condition number, or the mean-field marginal-variance ratios, which depend only on $X^\top X$, σ , and τ . Under the controlled design above, the posterior precision matrix has the form

$$\Lambda_{\text{post}} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}, \quad a = \frac{n}{\sigma^2} + \frac{1}{\tau^2}, \quad b = \frac{n\rho_X}{\sigma^2}.$$

The eigenvalues of Λ_{post} are therefore

$$L = a + b = \frac{n(1 + \rho_X)}{\sigma^2} + \frac{1}{\tau^2}, \quad m = a - b = \frac{n(1 - \rho_X)}{\sigma^2} + \frac{1}{\tau^2}.$$

Hence the posterior condition number is

$$\kappa_{\text{post}} = \frac{L}{m} = \frac{n(1 + \rho_X)/\sigma^2 + \tau^{-2}}{n(1 - \rho_X)/\sigma^2 + \tau^{-2}}.$$

This expression shows explicitly how design collinearity induces both posterior curvature anisotropy and dependence between the regression coefficients. As ρ_X approaches one, the likelihood becomes nearly flat along one direction. The prior prevents the smallest curvature from becoming zero by adding τ^{-2} to both eigenvalues. Thus, in Bayesian linear regression, posterior geometry is not a property of the likelihood alone: it is shaped by the interaction between collinearity and prior regularization.

Figure 3.3 reports κ_{post} as a function of ρ_X for the three values of τ . With a vague prior, the condition number is almost the likelihood condition number. With a more informative prior, the smallest curvature direction is regularized and the posterior condition number is substantially reduced. For example, at $\rho_X = 0.99$, the condition number is approximately 197 when $\tau = 10$, but approximately 17 when $\tau = 0.3$. Thus a moderately informative prior reduces the posterior conditioning by roughly one order of magnitude.

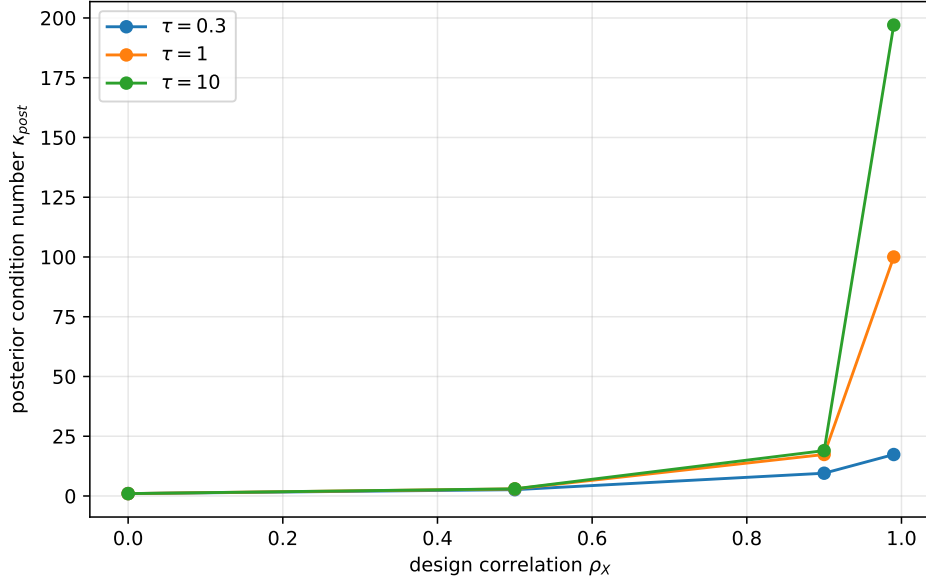


Figure 3.3: Posterior condition number κ_{post} as a function of design collinearity ρ_X , for the three prior scales $\tau \in \{0.3, 1, 10\}$.

In this Gaussian setting, κ_{post} plays a double role. On the sampling side, it directly controls the slow-direction autoregressive coefficient of ULA, since for $\eta = 0.8/L$,

$$1 - \eta m = 1 - \frac{0.8}{\kappa_{\text{post}}}.$$

On the variational side, the mean-field error is governed by the strength of posterior dependence, measured here by

$$|\text{corr}_\pi(\beta_1, \beta_2)| = \frac{b}{a} = \frac{\kappa_{\text{post}} - 1}{\kappa_{\text{post}} + 1}.$$

Thus, in the present symmetric Gaussian construction, the same collinearity parameter drives both posterior dependence and posterior conditioning. In general, however, dependence structure and curvature anisotropy are distinct geometric features and should not be identified.

3.3.2 Mean-field approximation and marginal uncertainty

We now study the optimal Gaussian mean-field approximation to the regression posterior. Let π denote the exact posterior distribution of β . We approximate π within the Gaussian mean-field family

$$q(\beta) = \mathcal{N}(r, D), \quad D = \text{diag}(d_1, d_2), \quad d_j > 0.$$

Here r denotes the variational mean parameter; the notation m is avoided in this section because m already denotes the smallest curvature eigenvalue $\lambda_{\min}(\Lambda_{\text{post}})$. As derived in Chapter 1 for a general Gaussian target, the minimizer of $KL(q||\pi)$ over this family is

$$q^*(\beta) = \mathcal{N}(\mu_{\text{post}}, D^*), \quad D^* = \text{diag}\left(\frac{1}{\Lambda_{\text{post},11}}, \frac{1}{\Lambda_{\text{post},22}}\right).$$

Thus the mean-field optimum preserves the posterior mean, but removes posterior dependence by construction. In the present symmetric design,

$$\Lambda_{\text{post},11} = \Lambda_{\text{post},22} = a,$$

so that

$$D^* = \frac{1}{a} I_2 = \frac{1}{n/\sigma^2 + \tau^{-2}} I_2.$$

The crucial point is that this mean-field marginal variance does not depend on ρ_X . Therefore, for fixed n, σ , and τ , the optimal mean-field approximation reports the same marginal uncertainty about β_1 whether the design is orthogonal or nearly collinear.

The exact posterior marginal variance behaves differently. Since

$$\Sigma_{\text{post}} = \Lambda_{\text{post}}^{-1} = \frac{1}{a^2 - b^2} \begin{pmatrix} a & -b \\ -b & a \end{pmatrix},$$

we have

$$\mathrm{Var}_\pi(\beta_1) = \mathrm{Var}_\pi(\beta_2) = \frac{a}{a^2 - b^2}.$$

Consequently, the posterior correlation between the two coefficients is

$$\mathrm{corr}_\pi(\beta_1, \beta_2) = \frac{\mathrm{Cov}_\pi(\beta_1, \beta_2)}{\sqrt{\mathrm{Var}_\pi(\beta_1) \mathrm{Var}_\pi(\beta_2)}} = -\frac{b}{a}.$$

Thus positive design collinearity, corresponding to $b > 0$, induces negative posterior dependence between the regression coefficients.

As ρ_X increases, $b = n\rho_X/\sigma^2$ increases and the denominator $a^2 - b^2$ decreases. The exact posterior therefore reflects the fact that individual coefficients become less precisely identified when the two predictors are nearly collinear.

The ratio between the optimal mean-field marginal variance and the exact posterior marginal variance is

$$\frac{\mathrm{Var}_{q^*}(\beta_1)}{\mathrm{Var}_\pi(\beta_1)} = \frac{1/a}{a/(a^2 - b^2)} = 1 - \left(\frac{b}{a}\right)^2.$$

By symmetry, the same ratio holds for β_2 . This ratio is equal to one when the design is orthogonal, and it decreases as the design becomes more collinear. Thus the mean-field approximation increasingly underestimates marginal posterior uncertainty.

The same scalar also determines the forward-KL error of the optimal Gaussian mean-field approximation. Since the exact posterior and the mean-field optimum have the same mean,

$$\mathrm{KL}(q^* \parallel \pi) = -\frac{1}{2} \log \left(1 - \left(\frac{b}{a}\right)^2 \right) = -\frac{1}{2} \log \left(\frac{\mathrm{Var}_{q^*}(\beta_1)}{\mathrm{Var}_\pi(\beta_1)} \right).$$

Hence the variance ratio and the optimal KL divergence are not independent diagnostics: in this symmetric Gaussian setting, they express the same mean-field mismatch in different parametrizations.

Figure 3.4 illustrates this phenomenon for the representative case $\tau = 1$. When $\rho_X = 0$, the posterior covariance is diagonal and the mean-field approximation coincides with the

target. As ρ_X increases, the exact posterior contours become increasingly elongated and tilted, reflecting negative posterior dependence between the coefficients. This negative dependence is expected: when two predictors are positively collinear, an increase in β_1 can be partly compensated by a decrease in β_2 , leaving the fitted value $X\beta$ nearly unchanged. The mean-field approximation removes this dependence by construction and remains axis-aligned.

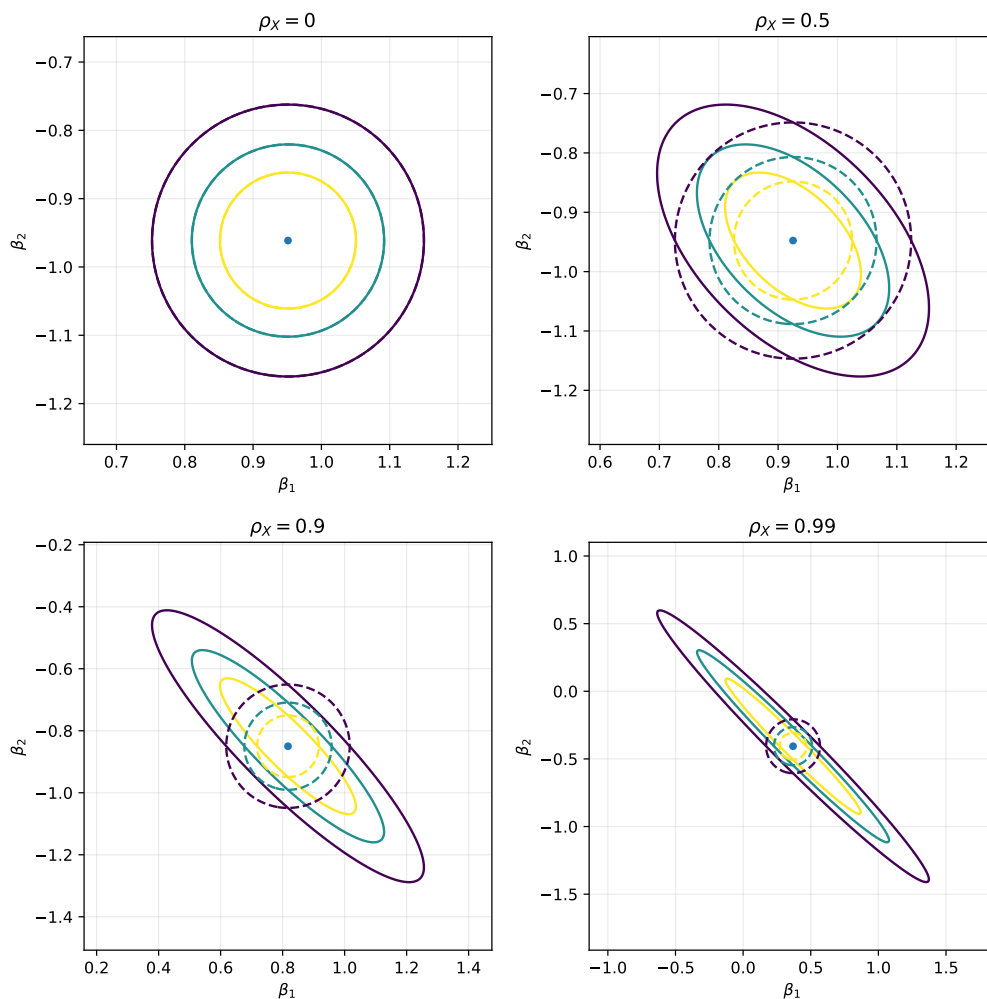


Figure 3.4: Exact posterior contours (solid) and optimal Gaussian mean-field approximations (dashed) for Bayesian linear regression with $\tau = 1$ and $\rho_X \in \{0, 0.5, 0.9, 0.99\}$.

Figure 3.5 reports the marginal-variance ratio as a function of ρ_X for the three prior scales. The effect is strongest under a vague prior and is attenuated by a more informative prior. This is consistent with the previous conditioning calculation: a more informative prior

increases curvature in the weakly identified direction and therefore reduces the posterior dependence that the factorized approximation fails to represent.

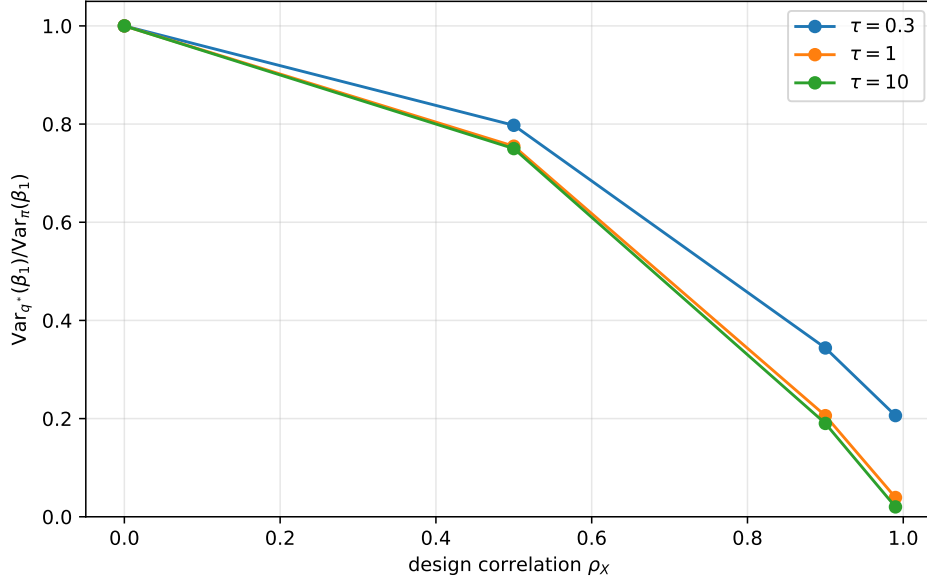


Figure 3.5: Ratio between the optimal mean-field and the exact posterior marginal variance of β_1 , as a function of ρ_X for the three prior scales. Values below one indicate mean-field underestimation.

Table 3.3 quantifies the magnitude of this effect for the representative case $\tau = 1$. The optimal mean-field variance remains constant across collinearity levels, while the exact posterior variance increases sharply as ρ_X approaches one. In the nearly collinear case $\rho_X = 0.99$, this corresponds to an underestimation of marginal posterior variance by a factor of about 25. This is the regression analogue of the structural failure discussed in Chapter 1: the approximation is optimal within the factorized family, but the family itself cannot represent the compensating dependence between coefficients.

ρ_X	κ_{post}	$\text{corr}_\pi(\beta_1, \beta_2)$	$\text{Var}_\pi(\beta_1)$	$\text{Var}_{q^*}(\beta_1)$
0.00	1.00	0.00	0.00990	0.00990
0.50	2.96	-0.50	0.01312	0.00990
0.90	17.36	-0.89	0.04807	0.00990
0.99	100.00	-0.98	0.25250	0.00990

Table 3.3: Posterior geometry and mean-field marginal variance for $\tau = 1$.

3.3.3 ULA dynamics in the posterior eigenspace

Finally, we examine the behaviour of ULA on the same posterior. Since the posterior is Gaussian, the potential is quadratic:

$$U(\beta) = \frac{1}{2}(\beta - \mu_{\text{post}})^\top \Lambda_{\text{post}}(\beta - \mu_{\text{post}}) + \text{constant},$$

and hence

$$\nabla U(\beta) = \Lambda_{\text{post}}(\beta - \mu_{\text{post}}).$$

ULA is therefore

$$\beta_{k+1} = \beta_k - \eta \Lambda_{\text{post}}(\beta_k - \mu_{\text{post}}) + \sqrt{2\eta} \xi_k, \quad \xi_k \sim \mathcal{N}(0, I_2).$$

Rotating into the eigenbasis of Λ_{post} , each coordinate evolves as an independent autoregressive process:

$$Y_{k+1}^{(i)} = (1 - \eta \lambda_i) Y_k^{(i)} + \sqrt{2\eta} \zeta_k^{(i)}.$$

The slow direction corresponds to $m = \lambda_{\min}(\Lambda_{\text{post}})$. With $\eta = 0.8/L$, its autoregressive coefficient is

$$1 - \eta m = 1 - \frac{0.8}{\kappa_{\text{post}}}.$$

Thus the same weakly identified direction that causes posterior variance inflation also causes slow ULA mixing.

Figure 3.6 shows the empirical autocorrelation of ULA along the slow posterior eigendirection for the representative case $\tau = 1$, together with the corresponding AR(1) theoretical decay implied by the Gaussian eigenspace decomposition. The curves are plotted for increasing values of ρ_X . As collinearity increases, the posterior condition number increases and the autocorrelation decays more slowly. The same mechanism reappears here: ULA is not restricted by a factorized approximation family, but its exploration becomes slow along the low-curvature direction created by collinearity.

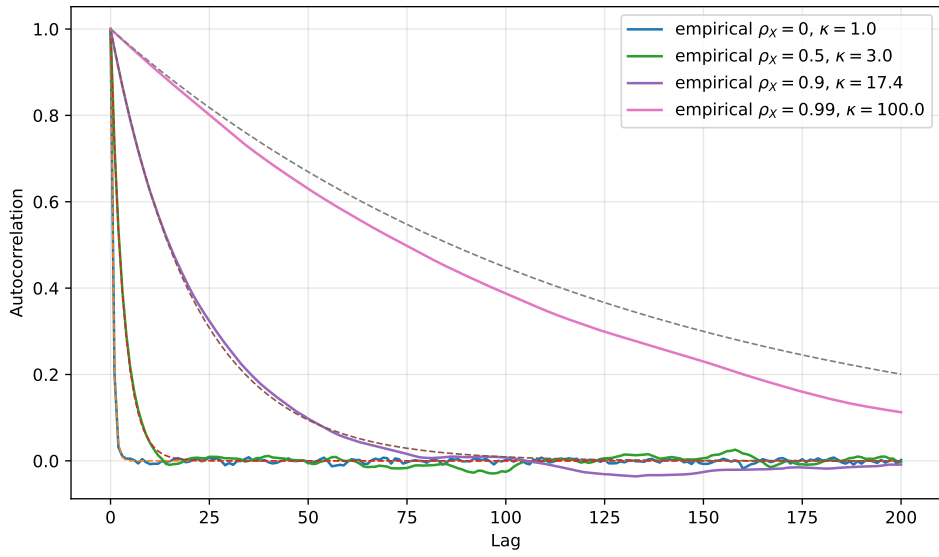


Figure 3.6: Empirical (solid) and theoretical AR(1) (dashed) autocorrelation of ULA along the slow posterior eigendirection, for $\tau = 1$ and $\rho_X \in \{0, 0.5, 0.9, 0.99\}$.

The experiment therefore complements the bivariate Gaussian example rather than duplicating it. In Experiment 1, the geometry was imposed directly through the target covariance. Here, the same type of geometry is generated by collinearity in a Bayesian regression model and modulated by the prior. The low-curvature direction now has a statistical interpretation: it corresponds to weak identification of individual coefficients.

Chapter 4

Discussion

4.1 The structural-dynamical asymmetry

Mean-field variational inference and the unadjusted Langevin algorithm fail along different axes when confronted with the same posterior geometry. The axis of failure is determined by where the approximation enters the inference procedure.

Mean-field variational inference commits to an approximation at the level of representation. Once a factorized family has been chosen, the procedure searches only within that family. If the posterior dependence structure lies outside it, no improvement in the optimization routine can recover the missing dependence. The gap between the variational optimum and the target is therefore a property of the family, not merely of the algorithm used to optimize the ELBO. Approximation here is structural: the error is encoded in the class of distributions allowed by the method. In the Gaussian examples studied above, this structural mismatch appears as variance shrinkage, removal of off-diagonal covariance, and a positive optimal KL divergence. The numerical analysis is exact; the answer it produces is approximate by design.

ULA introduces approximation at a later stage, and in a graded way. There is no restricted family: the continuous-time Langevin diffusion has the posterior itself as its invariant law, and in this sense the dynamics target the right object. The approximation appears only when the diffusion is replaced by an Euler scheme, and even then the error is graded by the

step size. Under the usual stability and regularity assumptions, smaller step sizes reduce the discretization error and make the long-run law of ULA closer to the target. The cost is paid in mixing time rather than in expressiveness: directions of high curvature constrain the step size, and once the step size is constrained, directions of low curvature are explored slowly. The error here is dynamical, in the precise sense that it is the trajectory of the algorithm, not the object it is aiming at, that is approximate.

The two failure modes therefore originate in the same geometric input but enter the two procedures at different stages. Posterior dependence is what the variational route refuses by construction, so it accumulates as approximation error. Posterior anisotropy is what the sampling route must respect in the steep and the flat directions at once, so it accumulates as mixing cost. In the Gaussian case the two phenomena are linked: the covariance and precision matrices share eigendirections with reciprocal eigenvalues, so covariance geometry, curvature, and conditioning are tied to the same linear-algebraic structure. This coincidence is the analytic anchor on which the comparison rests, and it is what allows the two failure modes to be exhibited exactly and simultaneously in a single target. The link is a feature of the Gaussian setting, not a general identity: dependence structure and curvature anisotropy remain distinct features of a posterior, and in non-Gaussian targets they can move independently.

This is what makes posterior geometry a useful diagnostic rather than a single scalar of difficulty. Asking whether the obstacle is dependence or anisotropy is asking whether the obstacle will manifest as a limitation of representation or a limitation of dynamics, and the two limitations are not interchangeable. A representational limit cannot be alleviated by running longer; a dynamical limit cannot be removed by optimizing the variational objective more carefully. Each method’s response to the same posterior is shaped by what it has already given up in exchange for tractability.

4.2 Beyond the two prototypes

The comparison developed in this thesis focuses on two deliberately simple prototypes: fully factorized mean-field variational inference and the unadjusted Langevin algorithm. This choice makes the two failure modes analytically visible, but it should not be interpreted as a statement about the limits of variational inference or MCMC as broad methodological classes. Both sides admit natural refinements that modify, but do not erase, the role of posterior geometry.

Richer variational families. On the variational side, the most direct response to the structural limitation of mean-field VI is to enlarge the variational family. Normalizing flows (Rezende and Mohamed, 2015) provide one such route: instead of approximating the posterior by an independent product distribution, they transform a simple base density through a sequence of invertible maps, producing a more flexible approximate distribution. This directly addresses the representational failure of mean-field VI, since a sufficiently flexible transformation can encode dependence structures that are excluded by a fully factorized family. The additional flexibility, however, comes with a larger parametrization, a more difficult optimization problem, and dependence on the chosen flow architecture. Normalizing flows therefore soften the structural limitation of mean-field VI, but they replace it with a different trade-off between expressiveness, computation, and optimization.

Improved sampling dynamics. On the sampling side, ULA can also be improved without abandoning the Langevin perspective. The discretization bias of ULA is specific to the unadjusted Euler scheme. Metropolis-adjusted Langevin algorithms (Roberts and Tweedie, 1996) correct this bias by using the Langevin proposal inside a Metropolis–Hastings acceptance step, thereby restoring exact π -invariance. This changes the nature of the cost: the algorithm no longer has the same invariant-measure bias as ULA, but it introduces accept/reject behaviour and additional sensitivity to proposal tuning. Hamiltonian Monte Carlo (Neal, 2011) goes further by using momentum-augmented dynamics to

propose longer-distance moves and reduce random-walk behaviour. These methods show that the dynamical failure mode of ULA is not an unavoidable property of all MCMC: it is the visible cost of the simplest unadjusted gradient-based discretization. MALA and HMC modify that cost, but they still interact with posterior geometry through step sizes, acceptance probabilities, integration times, and curvature of the target.

Coordinate-wise geometry and dependence-aware conditioning. The previous two extensions refine one of the two prototypes while preserving the contrast between variational representability and sampling dynamics. A different direction is suggested by coordinate-wise methods, where the relevant geometry may no longer be the global curvature condition number $\kappa = L/m$. This is important because mean-field VI is itself a product-space approximation, and its standard optimization algorithm, coordinate ascent variational inference, updates one factor at a time. On the sampling side, the analogous coordinate-wise method is the Gibbs sampler, which updates one coordinate or block from its conditional distribution. Coordinate descent, CAVI, and Gibbs therefore form a different comparison axis: instead of comparing a factorized variational family with a gradient-based sampler, one compares coordinate-wise optimization, coordinate-wise variational approximation, and coordinate-wise sampling.

For gradient-based methods such as ULA, the standard log-concave theory is naturally expressed in terms of m , L , and $\kappa = L/m$. For coordinate-wise methods, however, global anisotropy is not always the most informative measure. Recent work on Gibbs sampling under log-concavity (Ascolani et al., 2026) shows that, for strongly log-concave targets, the random-scan Gibbs sampler contracts in relative entropy, with the contraction rate governed by a dependence-aware constant κ_* , satisfying $\kappa_* \leq \kappa$, rather than only by the global condition number κ . Coordinate descent, CAVI, and Gibbs can then be placed in a common conceptual picture, in which coordinate-wise rates depend on both a dimension factor and a geometry-dependent conditioning constant. In this line of work, rates of the form $1 - 1/(\kappa_* d)$ arise for suitable random-scan coordinate-wise schemes, suggesting

a two-by-two landscape: gradient-wise versus coordinate-wise methods on one axis, and optimization or variational approximation versus sampling on the other.

This coordinate-wise viewpoint does not change the main conclusion of the thesis; it refines its scope. Curvature anisotropy is the natural quantity for gradient-based samplers, and dependence structure is the central obstruction for factorized variational approximations. Coordinate-wise methods indicate that there are settings where dependence, rather than spectral spread alone, is the more relevant notion of geometry. This suggests a broader research direction: posterior geometry is not a single scalar property of the target, but a method-dependent object. The geometry that limits a method depends on how that method interacts with the target: through a restricted approximating family, through a gradient-driven dynamics, or through coordinate-wise conditional updates (Arnese and Lacker, 2024; Ascolani et al., 2026).

Conclusion

This thesis has examined posterior geometry as a way to understand the behaviour of approximate Bayesian inference methods. The main point is not that one method is uniformly preferable to another, but that different methods are sensitive to different geometric features of the same target distribution. Mean-field variational inference is limited by the dependence structures excluded by its factorized family, while ULA is limited by the dynamical consequences of curvature anisotropy and discretization.

The analysis has been deliberately centred on smooth, strongly log-concave targets, with the explicit calculations and experiments carried out in Gaussian settings. Within that scope, the analysis can be made as precise as controlled experiments require: the two mechanisms are derived in closed form and verified empirically against an exact ground truth. Outside that scope, additional phenomena enter that this work has not addressed. Multimodal posteriors introduce a different difficulty: simple mean-field approximations may concentrate on one mode or average modes poorly, while Langevin-type chains can take very long to cross low-probability barriers. Heavy tails violate the strong log-concavity bounds on which the Wasserstein contraction results rest, and may change or preclude the contraction rates available in that framework. Non-log-concave geometries no longer admit a global curvature condition number κ in the sense used here, and the comparison between dependence structure and curvature anisotropy would need to be reformulated in their setting. The conclusions of this thesis should therefore be read as a precise statement about a particular regime, not as a general ranking of variational

inference and Langevin sampling.

What the regime exposes, however, is a usable principle. Posterior geometry separates a structural approximation failure from a dynamical-computational failure, and the separation is itself a guide to method choice. When the relevant geometric feature of a posterior is dependence, hence when coordinates carry information that mean-field VI would by construction discard, a method whose family encodes that dependence is worth its additional cost. When the relevant feature is anisotropy, so when curvature varies by orders of magnitude across directions, a sampler whose dynamics respect that anisotropy, through preconditioning, momentum, or coordinate-wise updates, is the priority. Knowing which feature dominates is what posterior geometry gives. In this sense, the thesis has not proposed a new algorithm; it has proposed a question to ask before choosing one.

Chapter A

Computational details

This appendix collects the computational conventions used to produce the experiments of Chapter 3. The aim is to make the numerical claims of the chapter fully reproducible, while keeping the body of the thesis free from implementation details that would distract from the mathematical content.

A.1 Software stack and reproducibility

All computations were performed in Python with `numpy`, `pandas`, and `matplotlib`. No probabilistic programming framework was used: every quantity reported in Chapter 3 either has a closed form (posterior mean and covariance, mean-field optimum, KL divergence, AR(1) decay rate) or is computed directly from explicit recursions (ULA, empirical autocorrelation, effective sample size).

Reproducibility is enforced by deterministic seeding of `numpy.random.default_rng`. The implementation uses integer sub-seeds. In Experiment 1 each value of ρ uses the sub-seed

$$s_\rho = s_0 + \lfloor 10000 \rho \rfloor, \quad s_0 = 123.$$

In Experiment 2 each (ρ_X, τ) cell uses the sub-seed

$$s_{\rho_X, \tau} = s_0 + \lfloor 10000 \rho_X \rfloor + \lfloor 1000 \tau \rfloor, \quad s_0 = 12345.$$

The noise vector ε entering $y = X\beta_{\text{true}} + \sigma\varepsilon$ is drawn once from the base RNG and reused across all (ρ_X, τ) cells, so that differences across cells are attributable to the design and the prior rather than to noise realisations. Posterior covariance, mean-field optimum, and the KL divergence depend only on $X^\top X$, σ and τ , and are therefore unaffected by the noise realisation.

A.2 Algorithmic conventions

ULA recursion. The unadjusted Langevin algorithm is implemented as the explicit Euler–Maruyama discretization of the Langevin SDE introduced in Chapter 2. For a Gaussian target with potential $U(x) = \frac{1}{2}(x - \mu)^\top \Lambda(x - \mu)$ the recursion reduces to

$$x_{k+1} = x_k - \eta \Lambda(x_k - \mu) + \sqrt{2\eta} \xi_k, \quad \xi_k \sim \mathcal{N}(0, I_d).$$

The step size is taken at the natural stability scale $\eta = c/L$ with $L = \lambda_{\max}(\Lambda)$ and $c = 0.8$. This choice keeps the chain inside the stability region of the discretized Gaussian process while preserving the dependence of the slow-direction contraction rate on the condition number $\kappa = L/m$, as discussed in Chapter 2.

Initialisation and burn-in. In Experiment 1 the chain is started at the posterior mean and the first 10,000 iterations are discarded; the remaining 40,000 iterations are used for diagnostics. In Experiment 2 the chain is initialised at the posterior mean, run for 60,000 iterations, and the first 10,000 are discarded. Starting at the posterior mean reduces transient bias in stationary-regime diagnostics and isolates the geometric slowdown from initial-condition effects; convergence from overdispersed initial conditions is not the object of this experiment.

Empirical autocorrelation. For a centred chain $z = (z_0, \dots, z_{N-1})$ the biased estimator of the lag- ℓ autocorrelation is

$$\hat{\rho}(\ell) = \frac{\sum_{k=0}^{N-1-\ell} z_k z_{k+\ell}}{\sum_{k=0}^{N-1} z_k^2}, \quad \ell = 0, 1, \dots, L_{\max}.$$

This estimator is used throughout, with $L_{\max} = 200$ in both experiments.

Effective sample size. The ESS is estimated by the initial-positive sequence rule of Geyer (Geyer, 1992):

$$\widehat{\text{ESS}} = \frac{N}{1 + 2 \sum_{\ell=1}^{L^*} \hat{\rho}(\ell)}, \quad L^* = \min\{\ell \geq 1 : \hat{\rho}(\ell) \leq 0\} - 1.$$

This is the same rule used by standard MCMC diagnostics packages and was chosen for its simplicity and absence of tunable parameters.

Slow and fast directions. For each Gaussian target the precision matrix is diagonalized, $\Lambda = Q \text{diag}(\lambda_1, \dots, \lambda_d) Q^\top$ with $\lambda_1 \leq \dots \leq \lambda_d$. The eigenbasis decouples the ULA recursion into independent AR(1) processes. The *slow direction* is the first eigenvector (smallest curvature $m = \lambda_1$) and the *fast direction* is the last (largest curvature $L = \lambda_d$). The empirical chain $\beta_k - \mu$ is projected onto these directions before computing $\hat{\rho}$ and ESS.

A.3 Experiment 1: bivariate Gaussian target

The target distributions are $\pi_\rho = \mathcal{N}(0, \Sigma_\rho)$, $\Sigma_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, for $\rho \in \{0, 0.5, 0.9, 0.99\}$. The optimal Gaussian mean-field optimum is $q_\rho^* = \mathcal{N}(0, (1 - \rho^2)I_2)$ and the closed-form KL is $\text{KL}(q_\rho^* \parallel \pi_\rho) = -\frac{1}{2} \log(1 - \rho^2)$. The ULA recursion uses 50,000 iterations with 10,000 burn-in and the step rule $\eta = 0.8/L$. The core of the simulation is the following.

```

1 def simulate_ula_gaussian(rho, n_steps, burn_in, c, rng):
2     Sigma = np.array([[1.0, rho], [rho, 1.0]])
3     Lambda = np.linalg.inv(Sigma)
```

```

4     eigvals, eigvecs = np.linalg.eigh(Lambda)
5     L = float(eigvals[-1])
6     eta = c / L
7     sqrt_2eta = np.sqrt(2.0 * eta)
8     x = np.zeros(2)
9     chain = np.empty((n_steps, 2))
10    for k in range(n_steps):
11        x = x - eta * (Lambda @ x) + sqrt_2eta * rng.normal(size=2)
12        chain[k] = x
13    return chain[burn_in:], eigvals, eigvecs, eta

```

After burn-in the chain is rotated into the eigenbasis of Σ_ρ^{-1} , and the empirical ACF and ESS of the slow component are computed via the estimators defined in Section A.2. The four reported quantities (κ , η , MF variance, $\text{KL}(q_\rho^* \parallel \pi_\rho)$) used in Table 3.1 are analytic; the ULA diagnostics in Table 3.2 are empirical.

A.4 Experiment 2: Bayesian linear regression

The model is $y \mid \beta \sim \mathcal{N}(X\beta, \sigma^2 I_n)$, $\beta \sim \mathcal{N}(0, \tau^2 I_2)$, with $n = 100$, $\sigma = 1$, $\tau \in \{10, 1, 0.3\}$, and $\rho_X \in \{0, 0.5, 0.9, 0.99\}$. The design matrix is built deterministically so that $X^\top X = n \begin{pmatrix} 1 & \rho_X \\ \rho_X & 1 \end{pmatrix}$ exactly, which removes Monte Carlo variability from the posterior geometry and makes the dependence of every reported quantity on (ρ_X, τ) analytically explicit. The closed-form posterior and the closed-form mean-field optimum are implemented as

```

1 def posterior_from_regression(X, y, sigma, tau):
2     p = X.shape[1]
3     precision = (X.T @ X) / sigma**2 + np.eye(p) / tau**2
4     covariance = np.linalg.inv(precision)
5     mean = covariance @ (X.T @ y / sigma**2)
6     return mean, covariance, precision
7

```

```

8 def gaussian_mf_forward_kl_optimum(mean, precision):
9     d_mf = 1.0 / np.diag(precision)
10    return mean.copy(), np.diag(d_mf)

```

ULA on the regression posterior follows the same Euler–Maruyama recursion as in Experiment 1, with the regression precision Λ_{post} in place of Σ_ρ^{-1} :

```

1 def simulate_ula_gaussian(mean, precision, eta, iterations, burnin, rng):
2     d = mean.shape[0]
3     x = mean.copy()
4     samples = np.empty((iterations - burnin, d))
5     sqrt_2eta = np.sqrt(2.0 * eta)
6     out = 0
7     for k in range(iterations):
8         x = x - eta * (precision @ (x - mean)) + sqrt_2eta * rng.normal(size=d)
9         if k >= burnin:
10             samples[out] = x
11             out += 1
12    return samples

```

The KL divergence between the optimal Gaussian mean-field and the exact posterior uses the closed-form Gaussian KL,

$$\text{KL}(q\|p) = \frac{1}{2} \left[\text{tr}(\Sigma_p^{-1} \Sigma_q) + (\mu_p - \mu_q)^\top \Sigma_p^{-1} (\mu_p - \mu_q) - d + \log \frac{\det \Sigma_p}{\det \Sigma_q} \right],$$

specialised in the body of the chapter to $\text{KL}(q^*\|\pi) = -\frac{1}{2} \log(1 - (b/a)^2)$ in the symmetric design.

Bibliography

- Arnese, M. and Lacker, D. (2024). “Convergence of coordinate ascent variational inference for log-concave measures via optimal transport”. arXiv preprint. Available at <https://arxiv.org/abs/2404.08792> [last accessed: 08/06/2026].
- Ascolani, F., Lavenant, H., and Zanella, G. (2026). “Entropy contraction of the Gibbs sampler under log-concavity”. *Annals of Probability*. In press. Preprint available at <https://arxiv.org/abs/2410.00858> [last accessed: 08/06/2026].
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer, New York.
- Blei, D. M., Kucukelbir, A., and McAuliffe, J. D. (2017). “Variational inference: a review for statisticians”. *Journal of the American Statistical Association*, 112(518):859–877.
- Chewi, S. (2026). Log-concave sampling. Book draft, version of June 5, 2026. Available at <https://chewisinho.github.io/main.pdf> [last accessed: 08/06/2026].
- Dalalyan, A. (2017a). “Further and stronger analogy between sampling and optimization: Langevin Monte Carlo and gradient descent”. In Kale, S. and Shamir, O., editors, *Proceedings of the 2017 Conference on Learning Theory*, volume 65 of *Proceedings of Machine Learning Research*, pages 678–689. PMLR.
- Dalalyan, A. S. (2017b). “Theoretical guarantees for approximate sampling from smooth and log-concave densities”. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 79(3):651–676.

- Durmus, A. and Moulines, É. (2016). “High-dimensional Bayesian inference via the unadjusted Langevin algorithm”. arXiv preprint. Available at <https://arxiv.org/abs/1605.01559> [last accessed: 08/06/2026].
- Geyer, C. J. (1992). “Practical Markov chain Monte Carlo”. *Statistical Science*, 7(4):473–483.
- Neal, R. M. (2011). “MCMC using Hamiltonian dynamics”. In Brooks, S., Gelman, A., Jones, G. L., and Meng, X.-L., editors, *Handbook of Markov Chain Monte Carlo*, pages 113–162. Chapman and Hall/CRC, Boca Raton, FL.
- Rezende, D. J. and Mohamed, S. (2015). “Variational inference with normalizing flows”. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1530–1538. PMLR.
- Roberts, G. O. and Tweedie, R. L. (1996). “Exponential convergence of Langevin distributions and their discrete approximations”. *Bernoulli*, 2(4):341–363.